Week 14: Governance, Standards and Auditing

Machine Ethics and Artificial Intelligence Safety

Machine Ethics and Artificial Intelligence Safety (11118BLG001). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Technical safeguards need institutions around them. This final week moves from ethics principles to binding law, international instruments, risk management frameworks and management system standards. It closes with documentation and auditing practices that make safety claims checkable [1, 2, 3, 4].

Learning outcomes

By the end of the week, students are expected to explain why principles alone cannot guarantee ethical AI. They should be able to describe the risk-based structure of the EU AI Act and the role of international instruments, including those relevant to Türkiye. They should also be able to apply the functions of the NIST AI Risk Management Framework. Finally, they should be able to produce a model card and an audit plan for a system.

Lecture

From principles to practice

Jobin, Ienca and Vayena analysed 84 documents containing AI ethics guidelines and found convergence around five principles: transparency, justice and fairness, non-maleficence, responsibility and privacy [1]. They also found substantial divergence in how the principles were interpreted and how they should be implemented. Floridi and Cowls proposed a unified framework of five principles for AI in society: beneficence, non-maleficence, autonomy, justice and explicability [5].

Mittelstadt argued that principles alone cannot guarantee ethical AI [6]. Many principles were borrowed from medicine, but AI development differs from medicine in four ways. It lacks common aims and fiduciary duties towards those affected, and a professional history with established norms. It also lacks proven methods to translate principles into practice and robust mechanisms of legal and professional accountability. The remaining sections of the week examine instruments that try to supply what principles lack.

Law and international instruments

The EU AI Act, Regulation (EU) 2024/1689, follows a risk-based approach [2]. A list of practices is prohibited, including social scoring, manipulation that exploits vulnerabilities, untargeted scraping of facial images and emotion recognition in workplaces and educational institutions, with narrow exceptions. High-risk systems include safety components of regulated products. They also include systems used in areas such as education, employment, access to essential services, law enforcement, migration and the administration of justice. Such systems must meet requirements on risk management, data governance, technical documentation and record-keeping. Further requirements concern transparency to deployers, human oversight, accuracy, robustness and cybersecurity. Systems that interact with people or generate synthetic content carry transparency obligations, and general-purpose AI models have their own obligations, with additional duties for models with systemic risk. The regulation entered into force on 1 August 2024. Its prohibitions applied from 2 February 2025 and the obligations for general-purpose models from 2 August 2025. The remaining obligations are phased in later. Because the later timetable has been subject to amendment proposals, the current consolidated text should be checked before any practical use. Figure 14.1 summarises the structure.

Other instruments complement the regulation. The Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law is the first legally binding international treaty on AI [7]. It opened for signature on 5 September 2024. The OECD Recommendation on AI, adopted in 2019 and amended in 2024, and the UNESCO Recommendation on the Ethics of AI, adopted in 2021, provide widely endorsed soft law [8, 9]. In Türkiye, the Digital Transformation Office of the Presidency and the Ministry of Industry and Technology prepared the National Artificial Intelligence Strategy 2021-2025 [10]. Its action plan was updated for 2024-2025.

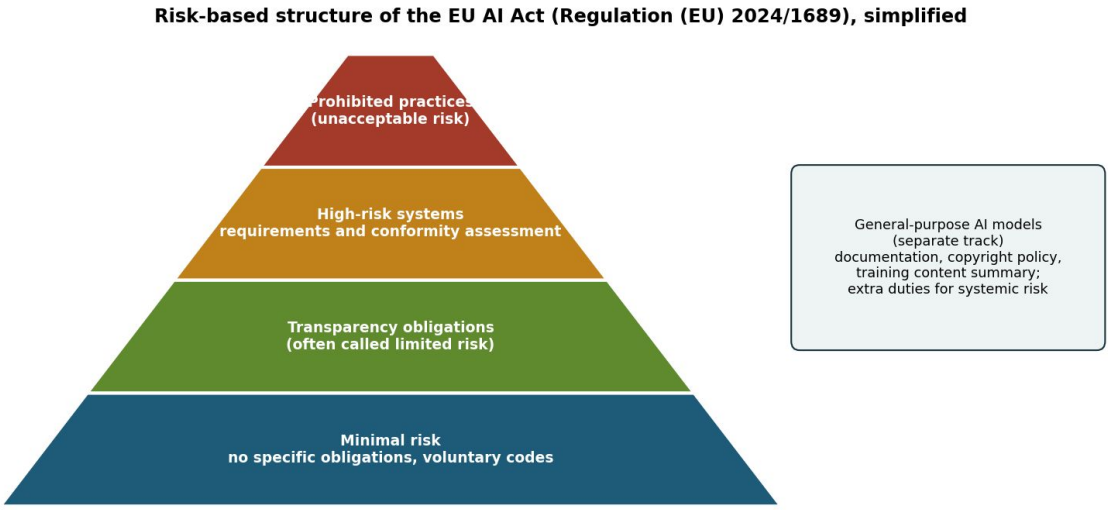


Figure 14.1. Simplified risk-based structure of the EU AI Act, with the separate track for general-purpose AI models. The figure is a teaching aid, not a legal summary [2].

Risk management and management systems

The NIST AI Risk Management Framework organises activities into four functions [3]. Govern establishes a culture, policies and accountability for AI risk. Map establishes the context and identifies risks. Measure analyses and tracks them with quantitative and qualitative methods. Manage prioritises risks and acts on them. The framework also describes the characteristics of trustworthy AI. They include validity and reliability, safety, security and resilience, accountability and transparency, explainability and interpretability, privacy enhancement and fairness with harmful bias managed. Each earlier week of the course supplies methods for the Measure function.

ISO/IEC 42001 specifies requirements for an AI management system, with which an organisation establishes, implements, maintains and continually improves its governance of AI, and conformity can be certified [11]. At the level of global science policy, the International AI Safety Report synthesised the evidence on the capabilities and risks of general-purpose AI for policymakers [12].

Documentation and auditing

Documentation makes claims checkable. Model cards report a model's intended use, the factors and metrics used for evaluation, performance across relevant groups and conditions, ethical considerations and caveats [13]. Datasheets for datasets document the motivation, composition, collection process, preprocessing, uses, distribution and maintenance of a data set [14]. Weidinger and colleagues provided a taxonomy of risks posed by language models, which helps auditors decide what to test [15].

Raji and colleagues proposed SMACTR, an end-to-end framework for internal algorithmic auditing with five stages: scoping, mapping, artifact collection, testing and reflection [4]. Mökander and colleagues proposed a three-layered approach for large language models: governance audits of the providers, model audits before release and application audits of the downstream uses [16]. A safety case is a natural way to assemble the results of such audits. It is a structured argument, supported by evidence, that a system is acceptably safe for a specific use in a specific context. The final task of the course asks for one.

> ***Pause and reflect.*** *Return to the question written down in Week 1. Has the course answered it, and which governance instrument from this week would matter most for the system it concerns?*

Hands-on components

Interactive lab: Part A walks through simplified questions about the EU AI Act and lists the obligations that may apply. It is a teaching aid, not legal advice. Part B drafts a model card in the structure of Mitchell and colleagues and exports it as a Markdown file [2, 13]. Link: https://utkukose.github.io/machine-ethics-ai-safety-NB-lecture/weeks/week-14/lab.html

Colab notebook: It runs a small audit of the lending model from Week 5: performance, group metrics and robustness to input noise. It then generates a model card from the results and builds a risk register mapped to the NIST AI RMF functions [3, 4, 13]. Link: https://colab.research.google.com/github/utkukose/machine-ethics-ai-safety-NB-lecture/blob/main/weeks/week-14/NB14_governance.ipynb

Reflection prompts

1. Which obligations from Part A of the interactive lab would apply to a system in your field, and who in your organisation would carry them?
2. Which of Mittelstadt's missing elements is most absent in your field, and what could replace it?
3. Revisit your question from Week 1. Write your answer as it stands now and note what remains open.

Weekly task and submission

What to do. Prepare the governance documents for one AI system from your field. First, answer the simplified questions of the interactive lab and list the obligations that may apply under the EU AI Act. Second, write a model card for the system. Third, build a risk register of at least six risks with likelihood, impact and score, and map each risk to a function of the NIST AI RMF. Finally, name the weakest piece of evidence in your documents.

Where to look. Lecture: sections "Law and international instruments", "Risk management and management systems" and "Documentation and auditing". Interactive lab: Part A for the classification and Part B for the model card. Colab notebook: Section 3 for generating a model card and Section 4 for the risk register.

Example. A tool that screens job applications is a high-risk system under the EU AI Act, because employment is an Annex III area. Its risk register could include indirect discrimination against older applicants, with likelihood 3 and impact 5, mapped to the Measure function.

What to submit. The model card, the risk register and a note of about 300 words, together with the Colab notebook with Exercises 1 and 2 completed. The note lists the obligations under the EU AI Act and names the weakest evidence. This weekly task is separate from the final capstone, whose page lists its own deliverables.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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